

Section 9.1.

Exercise 9.1.13.

Solution.

Let (R, Θ) be the representation of (X, Y) in polar coordinates. Let $L : \mathbb{R}^+ \times [0, 2\pi) \rightarrow \mathbb{R}^2$ be the map defined by $L(r, \theta) = (r \cos(\theta), r \sin(\theta))$. For any $A \in \mathcal{B}_{\mathbb{R}^+ \times [0, 2\pi)}$,

$$\begin{aligned} \mathbb{P}((R, \Theta) \in A) &= \mathbb{P}(L(R, \Theta) \in L(A)) = \mathbb{P}((X, Y) \in L(A)) \\ &= \int_{L(A)} \frac{1}{\pi} \mathbb{1}_{\{x^2 + y^2 \leq 1\}} d\lambda(x, y) = \int_A \frac{1}{\pi} \mathbb{1}_{\{r^2 \leq 1\}} \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| d\lambda(r, \theta) \\ &= \int_A \frac{r}{\pi} \mathbb{1}_{\{r \leq 1\}} d\lambda(r, \theta). \end{aligned}$$

So the density of (R, Θ) is $\frac{r}{\pi} \mathbb{1}_{\{r \leq 1\}}$. Now define $Z = \mathbb{1}_{\{Y \in A\}}$. Let $B \in \mathcal{B}_{[0, 2\pi)}$, then by definition we have

$$\begin{aligned} \mathbb{E}(\mathbb{E}(Z|\Theta); \Theta \in B) &= \mathbb{P}(Y \in A, \Theta \in B) \\ &= \mathbb{P}(R \sin(\Theta) \in A, \Theta \in B) \\ &= \int_B \int_{[0, 1]} \frac{r}{\pi} \mathbb{1}_A(r \sin(\theta)) dr d\theta \\ &= \int_B p(1, \theta) d\theta, \end{aligned}$$

where $p(z, \theta)$ is the density of (Z, Θ) . Then by **Exercise 9.1.10.**, the conditional expectation is computed as $g(\Theta)$, where

$$g(\theta) = \frac{p(1, \theta)}{p(1, \theta) + p(0, \theta)} = 2 \int_{[0, 1]} r \mathbb{1}_A(r \sin(\theta)) dr.$$

Hence $\mathbb{P}(Y \in A|\Theta) = 2 \int_0^1 r \mathbb{1}_A(r \sin(\Theta)) dr$. □

Section 9.2.

Exercise 9.2.2.

If it's almost sure convergence in the statement, then it's not true in general.

Clarify. (Not converge almost surely) An example is sufficient. Let $([0, 1], \mathcal{B}_{[0, 1]}, \mathbb{P})$ be the probability space. Define random variables X_n as follows:

$$X_n = \mathbb{1}_{\left[\frac{k}{2^m}, \frac{k+1}{2^m}\right]} \quad \text{if } n = 2^m + k, k = 0, \dots, 2^m - 1.$$

Let $X = 0$. Then $\|X_n - X\|_{L^1} = \|X_n\|_{L^1} = 2^{-\lfloor \log_2(n) \rfloor} \rightarrow 0$ as $n \rightarrow \infty$, but $\{\omega \in [0, 1] : \lim_{n \rightarrow \infty} X_n(\omega) = 0\} = \emptyset$. Now take $\mathcal{G} = \mathcal{B}_{[0, 1]}$. Of course X_n is $\mathcal{G}$ -measurable and thus $\mathbb{E}(X_n|\mathcal{G}) = X_n$ a.s. Hence we've shown that $X_n \rightarrow X$ in L^1 but $\mathbb{E}(X_n|\mathcal{G})$ doesn't converge to $\mathbb{E}(X|\mathcal{G})$ almost surely.

Solution. (Convergence in L^1) Suppose $X_n \rightarrow X$ in L^1 . Then

$$\lim_{n \rightarrow \infty} \int_{\Omega} |X_n - X| d\mathbb{P} = \lim_{n \rightarrow \infty} \int_{\Omega} \mathbb{E}(|X_n - X| | \mathcal{G}) d\mathbb{P} = 0.$$

By Jensen's inequality for conditional expectation, $\mathbb{E}(|X_n - X| | \mathcal{G}) \geq |\mathbb{E}(X_n - X | \mathcal{G})|$ a.e. for all $n \in \mathbb{N}$. Hence

$$\int_{\Omega} \mathbb{E}(|X_n - X| | \mathcal{G}) d\mathbb{P} \geq \int_{\Omega} |\mathbb{E}(X_n - X | \mathcal{G})| d\mathbb{P} \text{ for all } n \in \mathbb{N}.$$

Therefore,

$$0 = \lim_{n \rightarrow \infty} \int_{\Omega} \mathbb{E}(|X_n - X| | \mathcal{G}) d\mathbb{P} \geq \limsup_{n \rightarrow \infty} \int_{\Omega} |\mathbb{E}(X_n - X | \mathcal{G})| d\mathbb{P}.$$

Also, since $\liminf_{n \rightarrow \infty} \int_{\Omega} |\mathbb{E}(X_n - X | \mathcal{G})| d\mathbb{P} \geq 0$, we have

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \int_{\Omega} |\mathbb{E}(X_n - X | \mathcal{G})| d\mathbb{P} = \lim_{n \rightarrow \infty} \int_{\Omega} |\mathbb{E}(X_n | \mathcal{G}) - \mathbb{E}(X | \mathcal{G})| d\mathbb{P} \\ &= \lim_{n \rightarrow \infty} \|\mathbb{E}(X_n | \mathcal{G}) - \mathbb{E}(X | \mathcal{G})\|_{L^1}, \end{aligned}$$

by the linearity of the conditional expectation. Hence $\mathbb{E}(X_n | \mathcal{G}) \rightarrow \mathbb{E}(X | \mathcal{G})$ in L^1 . $\square$

Exercise 9.2.3.

Solution. Let $\Omega = [0, 1]$. Define a collection of subsets $\mathcal{E}$ as

$$\mathcal{E} = \{\{Y \in O\} : O \text{ is open interval in } \Omega\}.$$

Claim. Let h be a measurable function such that $\mathbb{E}(\mathbb{E}(X|Y); A) = \mathbb{E}(h(Y); A)$ for all $A \in \mathcal{E}$, then $\mathbb{E}(X|Y) = h(Y)$ a.s.

Proof of claim. First note that $\mathcal{E}$ is a π -system since it is closed under finite intersections, and $\sigma(\mathcal{E}) = \sigma(Y)$. (Check: The collection $\mathcal{G} = \{B \subset \Omega : \{Y \in B\} \in \sigma(\mathcal{E})\}$ forms a sigma algebra. Clearly $\{O : O \text{ is open interval in } \Omega\} \subset \mathcal{G}$, hence $\mathcal{B}_{[0,1]} \subset \mathcal{G}$. Therefore $\sigma(Y) \subset \sigma(\mathcal{E})$.)

Suppose h is a measurable function that satisfies the condition. Since $\mathbb{E}(X|Y)$ and $h(Y)$ are non-negative measurable functions, $\mathbb{E}(\mathbb{E}(X|Y); \cdot)$ and $\mathbb{E}(h(Y); \cdot)$ induce two measures on Ω . Moreover, these two measures agree on a π -system $\mathcal{E}$. Now by a result of Dynkin's π - λ theorem (**Theorem 1.3.6.**), we can know that $\mathbb{E}(\mathbb{E}(X|Y); \cdot)$ and $\mathbb{E}(h(Y); \cdot)$ agree on $\sigma(\mathcal{E})$, hence agree on $\sigma(Y)$. Therefore $h(Y) = \mathbb{E}(X|Y)$ a.s. $\square$

Given any open interval $(a, b) \in [0, 1]$. Fix $n \in \mathbb{N}$, then

$$\mathbb{E}(X; Y \in (a, b)) = \sum_{k=0}^{n-1} \int_{\frac{a+k}{n}}^{\frac{b+k}{n}} \omega d\omega = \sum_{k=0}^{n-1} \frac{(b+k)^2 - (a+k)^2}{2n^2}.$$

Now by the definition of Y , one can see that $\mathbb{P}(Y \leq y) = \sum_{k=0}^{n-1} \frac{y}{n} = y$ for $y \in [0, 1]$, hence

$Y \sim \text{Unif}([0, 1])$. Now define $h : \Omega \rightarrow [0, 1]$ as

$$h(y) = \sum_{k=0}^{n-1} \frac{y+k}{n^2}.$$

Then

$$\mathbb{E}(h(Y); Y \in (a, b)) = \int_a^b h(y) d\mu_Y(y) = \int_a^b \sum_{k=0}^{n-1} \frac{y+k}{n^2} dy = \sum_{k=0}^{n-1} \frac{(y+k)^2}{2n^2} \Big|_a^b,$$

which agrees on $\mathbb{E}(X; Y \in (a, b))$. Hence by the claim we have $\mathbb{E}(X|Y) = h(Y)$ a.s. $\square$

Exercise 9.2.4.

Solution. Since $\mathbb{E}(X_1; A) \leq \mathbb{E}(X_2; A)$ for all $A \in \mathcal{F}$, $\mathbb{E}(X_1; X_1 > X_2) \leq \mathbb{E}(X_2; X_1 > X_2)$. Then

$$\mathbb{E}(X_2; X_1 > X_2) - \mathbb{E}(X_1; X_1 > X_2) = \mathbb{E}(X_2 - X_1; X_1 - X_2 > 0) \geq 0.$$

This implies $\mathbb{P}(X_1 > X_2) = 0$. Hence $\mathbb{P}(X_1 \leq X_2) = 1$, $\square$

Exercise 9.2.5. (TBD)

Solution. $\mathbb{E}[(X - \mathbb{E}(X|\mathcal{G}))\mathbb{E}(Y|\mathcal{G})] = 0$; $\mathbb{E}[(Y - \mathbb{E}(Y|\mathcal{G}))\mathbb{E}(X|\mathcal{G})] = 0$.

Exercise 9.2.6. (TBD)

Solution.

Exercise 9.2.7.

Solution. Fix $a \in \mathbb{N}$. Define $A = \{|X| \leq a\}$. Then the random variables $X\mathbb{1}_A Y\mathbb{1}_{\{|Y| \leq b\}}$ is bounded by $a|Y|$ for all $b \in \mathbb{N}$. Since $\mathbb{E}(a|Y|) = a\mathbb{E}|Y| < \infty$ and $X\mathbb{1}_A Y\mathbb{1}_{\{|Y| \leq b\}} \rightarrow X\mathbb{1}_A Y$ as $b \rightarrow \infty$, by dominated convergence theorem we have

$$\begin{aligned} \lim_{b \rightarrow \infty} \mathbb{E}(X\mathbb{1}_A Y\mathbb{1}_{\{|Y| \leq b\}}) &= \mathbb{E}(X\mathbb{1}_A Y) \\ &= \mathbb{E}(\mathbb{E}(X\mathbb{1}_A Y|X)) \\ &= \mathbb{E}(X\mathbb{1}_A \mathbb{E}(Y|X)) \\ &= \mathbb{E}(X^2\mathbb{1}_A). \end{aligned}$$

Since $\mathbb{E}(X\mathbb{1}_A Y) \leq a\mathbb{E}|Y| < \infty$, $\mathbb{E}(X(X - Y); A) = 0$. Therefore,

$$\begin{aligned} \mathbb{E}(X(X - Y); |X| < \infty) &= \mathbb{E}\left(X(X - Y); \bigcup_{a \geq 1} \{|X| \leq a\}\right) \\ &= \sum_{a \geq 1} \mathbb{E}(X(X - Y); a < |X| \leq a + 1) \\ &\quad + \mathbb{E}(X(X - Y); |X| \leq 1) \\ &= 0, \end{aligned}$$

since $\mathbb{E}(X(X - Y); A) = 0$. Also note that $|X| < \infty$ a.s. since $\mathbb{E}|X| < \infty$, hence we have $\mathbb{E}(X(X - Y)) = 0$ a.s. Similarly one can prove $\mathbb{E}(Y(Y - X)) = 0$. Combine them together then we get $\mathbb{E}(X - Y)^2 = 0$, so $X - Y = 0$ a.s. $\square$

Exercise 9.2.8.

Solution. Let $U_t = \max\{S, t\}$. Then $\{U_t = t\} = \{S \leq t\}$; $\{U_t > t\} = \{S > t\}$. Define

$$g(U_t) = \frac{\mathbb{E}(S; S \leq t)}{\mathbb{P}(S \leq t)} \mathbb{1}_{\{U_t=t\}} + U_t \mathbb{1}_{\{U_t>t\}}.$$

Then $\mathbb{E}(g(U_t); U_t = t) = \mathbb{E}(S; S \leq t) = \mathbb{E}(S; U_t = t)$, and $\mathbb{E}(g(U_t); t < U_t < a) = \mathbb{E}(U_t; t < U_t < a) = \mathbb{E}(S; t < U_t < a)$. Hence $g(U_t) = \mathbb{E}(S|U_t)$ a.s. $\square$

Exercise 9.2.9.

Solution. (X, Y) can be wrote as a function of (Θ, Z) :

$$\begin{aligned} Z = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} &= \begin{pmatrix} \Theta & 1 - \Theta \\ 1 - \Theta & \Theta \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} X \\ Y \end{pmatrix} = \frac{1}{2\Theta - 1} \begin{pmatrix} \Theta & \Theta - 1 \\ \Theta - 1 & \Theta \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \\ &= L(\Theta, Z). \end{aligned}$$

Moreover, $Z \perp\!\!\!\perp \Theta$: Let μ be the law of (Z_1, Z_2) . Let $C, D \in \mathcal{B}_{\mathbb{R}}$. Then

$$\mathbb{P}(Z \in C \times D) = \{\Theta = 1, X \in C, Y \in D\} \cup \{\Theta = 0, X \in D, Y \in C\},$$

hence by X, Y, Θ are independent and X, Y have identical distribution,

$$\begin{aligned} \mathbb{P}(Z \in C \times D, \Theta = 1) &= p\mu(C)\mu(D) \\ &= p(p\mu(C)\mu(D) + (1 - p)\mu(D)\mu(C)) \\ &= \mathbb{P}(\Theta = 1)\mathbb{P}(Z \in C \times D). \end{aligned}$$

$\{\Theta = 0\}$ case is similar. Thus $Z \perp\!\!\!\perp \Theta$.

Now let $A \in \mathcal{B}_{\mathbb{R}^2}$. Then

$$\begin{aligned} \mathbb{E}(g(X, Y); Z \in A) &= \int_{\Omega} g(L(\Theta, Z)) \mathbb{1}_{\{Z \in A\}} d\mathbb{P} \\ &= \int_{\mathbb{R}^2} \sum_{\theta=0,1} g(L(\theta, \mathbf{z})) \nu(\theta) \mathbb{1}_A(\mathbf{z}) \mu(\mathbf{z}) d\mathbf{z}, \quad (Z \perp\!\!\!\perp \Theta) \end{aligned}$$

where ν is the law of Θ . Define h as follows,

$$h(\mathbf{z}) = \sum_{\theta=0,1} g(L(\theta, \mathbf{z})) \nu(\theta) = \mathbb{E}_{\Theta}((g \circ L)(\Theta, \mathbf{z})),$$

then $\mathbb{E}(g(X, Y); Z \in A) = \int_A h(\mathbf{z})\mu(\mathbf{z})d\mathbf{z}$. Hence

$$\mathbb{E}_\Theta((g \circ L)(\Theta, \mathbf{z}))\big|_{\mathbf{z}=Z} = \mathbb{E}(g(X, Y)|Z) \text{ a.s.}$$

□

Exercise 9.2.10.

Solution. Consider $B = \{\mathbb{E}(X_1|\mathcal{G}) \geq -b\}$. Then $\mathbb{E}(X_1|\mathcal{G})\mathbb{1}_B = \mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}) \geq -b$. Since $X_n \uparrow X$, for any $A \in \mathcal{G}$,

$$\mathbb{E}(\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}); A) = \mathbb{E}(X_1\mathbb{1}_B; A) \leq \mathbb{E}(X_2\mathbb{1}_B; A) = \mathbb{E}(\mathbb{E}(X_2\mathbb{1}_B|\mathcal{G}); A).$$

But if $A = \{\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}) > \mathbb{E}(X_2\mathbb{1}_B|\mathcal{G})\}$ then $\mathbb{E}(\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}); A) \geq \mathbb{E}(\mathbb{E}(X_2\mathbb{1}_B|\mathcal{G}); A)$. Hence $\mathbb{E}(\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}); A) = \mathbb{E}(\mathbb{E}(X_2\mathbb{1}_B|\mathcal{G}); A)$ a.s. This implies $A = \emptyset$, therefore $\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}) \leq \mathbb{E}(X_2\mathbb{1}_B|\mathcal{G})$. Thus $\{\mathbb{E}(X_n\mathbb{1}_B|\mathcal{G})\}_{n \in \mathbb{N}}$ is monotonically increasing. Now use monotone convergence theorem of the nonnegative functions (since $b + \mathbb{E}(X_n\mathbb{1}_B|\mathcal{G}) \geq 0$), for any $A \in \mathcal{G}$,

$$\int_A \lim_{n \rightarrow \infty} \mathbb{E}(X_n\mathbb{1}_B|\mathcal{G})d\mathbb{P} = \lim_{n \rightarrow \infty} \int_A \mathbb{E}(X_n\mathbb{1}_B|\mathcal{G})d\mathbb{P} = \lim_{n \rightarrow \infty} \int_A X_n\mathbb{1}_Bd\mathbb{P}.$$

Now we need to compute the last term. Since $0 \leq (X_n - X_1)\mathbb{1}_B \uparrow (X - X_1)\mathbb{1}_B$ a.s., by MCT again,

$$\lim_{n \rightarrow \infty} \int_A (X_n - X_1)\mathbb{1}_Bd\mathbb{P} = \int_A (X - X_1)\mathbb{1}_Bd\mathbb{P}.$$

Since $\mathbb{E}(X_1\mathbb{1}_B; A) = \mathbb{E}(\mathbb{E}(X_1\mathbb{1}_B|\mathcal{G}); A) \geq -b$, one can add this term in both sides. Then by the first equation,

$$\lim_{n \rightarrow \infty} \int_A X_n\mathbb{1}_Bd\mathbb{P} = \int_A \lim_{n \rightarrow \infty} \mathbb{E}(X_n\mathbb{1}_B|\mathcal{G})d\mathbb{P} = \int_A X\mathbb{1}_Bd\mathbb{P},$$

which is $\mathbb{E}(\mathbb{E}(X|\mathcal{G})\mathbb{1}_B; A)$. Since $A \in \mathcal{G}$ is arbitrary, by the first equation again, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n|\mathcal{G})\mathbb{1}_B = \mathbb{E}(X|\mathcal{G})\mathbb{1}_B \text{ a.s.}$$

Since $B = \{\mathbb{E}(X_1|\mathcal{G}) \geq -b\}$ increases to $\cup_{b \geq 1} \{\mathbb{E}(X_1|\mathcal{G}) \geq -b\} = \{\mathbb{E}(X_1|\mathcal{G}) > -\infty\}$,

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n|\mathcal{G})\mathbb{1}_{\{\mathbb{E}(X_1|\mathcal{G}) > -\infty\}} = \mathbb{E}(X|\mathcal{G})\mathbb{1}_{\{\mathbb{E}(X_1|\mathcal{G}) > -\infty\}} \text{ a.s.}$$

□

Exercise 9.2.11.

Solution. Fix $\epsilon > 0$. For each $n \in \mathbb{N}$, define X_n as

$$X_n = \sum_{k=0}^{n2^{n+1}-1} \left(-n + \frac{k}{2^n} \right) \mathbb{1}_{\{X \in [-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n})\}}.$$

Then $\mathbb{E}|X - X_n|$ can be computed as:

$$\begin{aligned}
\mathbb{E}|X - X_n| &= \mathbb{E} \left| \sum_{k=0}^{n2^{n+1}-1} \left(X + n - \frac{k}{2^n} \right) \mathbb{1}_{\{X \in [-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n})\}} + X \mathbb{1}_{\{X < -n, X \geq n\}} \right| \\
&\leq \mathbb{E} \left| \sum_{k=0}^{n2^{n+1}-1} \left(X + n - \frac{k}{2^n} \right) \mathbb{1}_{\{X \in [-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n})\}} \right| + \mathbb{E} \left| X \mathbb{1}_{\{X < -n, X \geq n\}} \right| \\
&\leq \sum_{k=0}^{n2^{n+1}-1} \mathbb{E} \left| \left(X + n - \frac{k}{2^n} \right) \mathbb{1}_{\{X \in [-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n})\}} \right| + \mathbb{E}(|X|; X < -n, X \geq n) \\
&\leq \sum_{k=0}^{n2^{n+1}-1} \frac{1}{2^n} \mathbb{P} \left(X \in \left[-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n} \right) \right) + \mathbb{E}(|X|; |X| \geq n) \\
&\leq \frac{1}{2^n} + \mathbb{E}(|X|; |X| \geq n).
\end{aligned}$$

One can verify that the second term in the RHS goes to 0 as $n \rightarrow \infty$: Since $\mathbb{E}|X| < \infty$, $|X| < \infty$ a.e., therefore $|X| \mathbb{1}_{\{|X| \geq n\}} \downarrow 0$ as $n \rightarrow \infty$ a.e. Also, since $|X| \mathbb{1}_{\{|X| \geq n\}} \leq |X|$ for all n , then by dominated convergence theorem we have $\lim_{n \rightarrow \infty} \mathbb{E}(|X|; |X| \geq n) = 0$.

So there exists $N \in \mathbb{N}$ such that $\mathbb{E}|X - X_n| < \epsilon$ whenever $n > N$. Now suppose $n > N$. Since $\cup_{i=1}^{\infty} \mathcal{F}_i$ is the algebra that generates $\mathcal{F}$, there exists $A_{k+1} \in \cup_{i=1}^{\infty} \mathcal{F}_i$, $k = 0, \dots, n2^{n+1} - 1$, such that

$$\mathbb{P} \left(A_{k+1} \Delta \left\{ X \in \left[-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n} \right) \right\} \right) < \frac{\epsilon}{n2^{k+1}}.$$

(See Theorem 1.2.6.)

Then define the simple function Y_n as

$$Y_n = \sum_{k=0}^{n2^{n+1}-1} \left(-n + \frac{k}{2^n} \right) \mathbb{1}_{A_{k+1}}.$$

Then

$$\begin{aligned}
\mathbb{E}|X_n - Y_n| &\leq \sum_{k=0}^{n2^{n+1}-1} \mathbb{E} \left| \left(-n + \frac{k}{2^n} \right) (\mathbb{1}_{A_{k+1}} - \mathbb{1}_{\{X \in [\frac{k}{2^n}, \frac{k+1}{2^n})\}}) \right| \\
&\leq \sum_{k=0}^{n2^{n+1}-1} n \mathbb{P} \left(A_{k+1} \Delta \left\{ X \in \left[-n + \frac{k}{2^n}, -n + \frac{k+1}{2^n} \right) \right\} \right) < \epsilon.
\end{aligned}$$

Therefore $\mathbb{E}|X - Y_n| \leq \mathbb{E}|X - X_n| + \mathbb{E}|X_n - Y_n| < 2\epsilon$. $\square$

Fix any $\epsilon > 0$ again. There exists a $\mathcal{F}$ -simple function Y_n , with sufficiently large n , such that $\mathbb{E}|X - Y_n| < \epsilon$. Since Y_n takes on finitely many sets in $\mathcal{F}$, there exists $N \in \mathbb{N}$ such that $A_1, \dots, A_{n2^{n+1}} \in \mathcal{F}_m$ for any $m > N$. Clearly Y_n is $\mathcal{F}_m$ -measurable. Therefore, if $m > N$,

$$\begin{aligned}
\mathbb{E}|\mathbb{E}(X|\mathcal{F}_m) - X| &\leq \mathbb{E}|\mathbb{E}(X|\mathcal{F}_m) - Y_n| + \mathbb{E}|Y_n - X| \\
&\leq \mathbb{E}(\mathbb{E}(|X - Y_n||\mathcal{F}_m)) + \mathbb{E}|Y_n - X| \\
&= 2\mathbb{E}|X - Y_n| < 2\epsilon.
\end{aligned}$$

Hence $\|\mathbb{E}(X|\mathcal{F}_m) - X\|_{L^1} \rightarrow 0$ as $m \rightarrow \infty$. □

Exercise 9.2.12.

Solution. Since $\mathbb{E}(Y) < \infty$, $Y < \infty$ a.s. Hence $Y\mathbb{1}_{\{Y \geq a\}} \downarrow 0$ as $a \rightarrow \infty$ a.s. Also, $|Y\mathbb{1}_{\{Y \geq a\}}| \leq |Y|$ for all $a \in \mathbb{N}$. Then $\lim_{a \rightarrow \infty} \mathbb{E}(Y; Y \geq a) = 0$ by dominated convergence theorem.

Fix $\epsilon, \delta > 0$. Set $a \in \mathbb{N}$ to be the integer such that $\mathbb{E}(Y; Y \geq a) < \frac{\epsilon\delta}{3}$.

Since $0 \leq X_n \rightarrow 0$ in probability, we can take $N \in \mathbb{N}$ s.t. $\mathbb{P}(X_n > \frac{\epsilon\delta}{3}) < \frac{\epsilon\delta}{3a}$ whenever $n > N$. Suppose $n > N$. Since $X_n \geq 0$ a.s., $\mathbb{E}(X_n|\mathcal{G}) \geq 0$ a.s. Then by Markov's inequality,

$$\begin{aligned} \mathbb{P}(\mathbb{E}(X_n|\mathcal{G}) > \epsilon) &< \frac{\mathbb{E}(\mathbb{E}(X_n|\mathcal{G}))}{\epsilon} = \frac{\mathbb{E}(X_n)}{\epsilon} \\ &= \frac{1}{\epsilon} \left(\mathbb{E}\left(X_n; X_n \leq \frac{\epsilon\delta}{3}\right) + \mathbb{E}\left(X_n; \frac{\epsilon\delta}{3} < X_n < a\right) + \mathbb{E}(X_n; X_n \geq a) \right) \\ &\leq \frac{1}{\epsilon} \left(\frac{\epsilon\delta}{3} + a\mathbb{P}\left(X_n > \frac{\epsilon\delta}{3}\right) + \mathbb{E}(X_n; X_n \geq a) \right) \\ &< \frac{1}{\epsilon} \left(\frac{\epsilon\delta}{3} + \frac{\epsilon\delta}{3} + \mathbb{E}(X_n; X_n \geq a) \right). \end{aligned}$$

Since $0 \leq X_n \leq Y$ a.s. for all $n \in \mathbb{N}$, $\mathbb{E}(X_n; X_n \geq a) \leq \mathbb{E}(Y; X_n \geq a) \leq \mathbb{E}(Y; Y \geq a)$. Therefore

$$\mathbb{P}(\mathbb{E}(X_n|\mathcal{G}) > \epsilon) < \frac{1}{\epsilon} \left(\frac{\epsilon\delta}{3} + \frac{\epsilon\delta}{3} + \mathbb{E}(Y; Y \geq a) \right) < \delta.$$

Hence $\mathbb{P}(\mathbb{E}(X_n|\mathcal{G}) > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$. □

Exercise 9.2.13.

Solution. Let $\mathcal{P} = \{A \cap B : A \in \mathcal{G}_1, B \in \mathcal{G}_2\}$. Then $\mathcal{P}$ is a π -system: Pick $E_1, E_2 \in \mathcal{P}$, then $E_1 = A_1 \cap B_1$ and $E_2 = A_2 \cap B_2$ for some $A_1, A_2 \in \mathcal{G}_1$ and $B_1, B_2 \in \mathcal{G}_2$. Clearly $E_1 \cap E_2 \in \mathcal{P}$. Moreover, $\mathcal{G}_1 \cup \mathcal{G}_2 \subset \mathcal{P} \subset \sigma(\mathcal{G}_1 \cup \mathcal{G}_2) = \mathcal{G} \subset \sigma(\mathcal{P})$. Now let $A \cap B \in \mathcal{P}$. Then

$$\begin{aligned} \mathbb{E}(\mathbb{E}(X|\mathcal{G}_1); A \cap B) &= \mathbb{E}(\mathbb{E}(X|\mathcal{G}_1)\mathbb{1}_A\mathbb{1}_B) \\ &= \mathbb{E}(\mathbb{E}(X|\mathcal{G}_1)\mathbb{1}_A)\mathbb{E}(\mathbb{1}_B) \quad (\mathcal{G}_1, \mathcal{G}_2 \text{ are independent}) \\ &= \mathbb{E}(X\mathbb{1}_A)\mathbb{P}(B) \quad (\text{conditional expectation on } \mathcal{G}_1) \\ &= \mathbb{E}(X\mathbb{1}_A\mathbb{1}_B) \quad (\sigma(\sigma(X) \cup \mathcal{G}_1), \mathcal{G}_2 \text{ are independent}) \\ &= \mathbb{E}(\mathbb{E}(X|\mathcal{G}); A \cap B). \quad (A \cap B \in \mathcal{G}) \end{aligned}$$

Hence $\mathbb{E}(\mathbb{E}(X|\mathcal{G}_1); \cdot) = \mathbb{E}(\mathbb{E}(X|\mathcal{G}); \cdot)$ agrees on $\mathcal{P}$. By a result of Dynkin's π - λ system, **Theorem 1.3.6.**, they agree on $\sigma(\mathcal{P})$ and hence on $\mathcal{G}$. Since $\mathbb{E}(X|\mathcal{G}_1)$ is of course a $\mathcal{G}$ -measurable function, by definition $\mathbb{E}(X|\mathcal{G}_1) = \mathbb{E}(X|\mathcal{G})$ a.s. □

Section 9.5.

Exercise 9.5.1.

Solution. $\{T_{a,b} = n\} = \bigcap_{i=1}^{n-1} \{a < S_i < b\} \cap \{S_n \in \{a, b\}\} \in \sigma(S_0, S_1, \dots, S_n) = \mathcal{F}_n$.

□

Exercise 9.5.2.

Solution. Define $T = \min\{n : S_k - S_{k-1} = 1 \text{ for all } k = n - (a + b) + 1, \dots, n\}$. Then $T_{a,b} < T$ a.s. And $\mathbb{E}T$ follows the following equation:

$$\mathbb{E}T = \sum_{k=0}^{a+b-1} p^k (1-p)(1+k+\mathbb{E}T) + p^{a+b}(a+b)$$

Exercise 9.5.3.

Solution. ($T < \infty$ a.s.) One can check that $\max\{n : S_n \geq n\} \leq N$ if and only if $S_k < k$ for all $k > N$. Hence $\{T \leq N\} = \bigcap_{k>N} \{S_k < k\}$. Since X_n are i.i.d. and $\mathbb{E}X_n = 0$, by SLLN we know that $S_n/n \rightarrow 0$ almost surely. Therefore, $\Omega = \bigcup_{N \geq 1} \bigcap_{k>N} \{|S_k|/k < 1\}$ a.s. Since $\bigcap_{k>N} \{|S_k|/k < 1\} \subset \{T \leq N\}$, $\Omega = \bigcup_{N \geq 1} \{T \leq N\} = \{T < \infty\}$ a.s.

(T is not a stopping time) Note that $\{T = 0\} = \bigcap_{k \geq 1} \{S_k < k\}$. Clearly $\{T = 0\} \neq \emptyset$, since the path $\omega = (0, -1, -2, -3, \dots) \in \bigcap_{k \geq 1} \{S_k < k\}$. Suppose $\{T = 0\} = \Omega$. Then $\{S_1 < 1\} = \{X_1 < 1\} = \Omega$. But $\mathbb{P}(X_1 < 1) = 1/2 \neq 1$, a contradiction. So $\{T = 0\} \notin \{\emptyset, \Omega\} = \mathcal{F}_0$, therefore T is not a stopping time.

□

Exercise 9.5.4.

Solution. Suppose T is a stopping time. Then $\{T \leq n\} = \{T > n\}^c \in \bigcup_{i=0}^n \mathcal{F}_i = \mathcal{F}_n$, so $\{T > n\} \in \mathcal{F}_n$. Suppose $\{T > n\} \in \mathcal{F}_n$ for all $n \geq 0$. Then $\{T \leq n+1\} \in \mathcal{F}_{n+1}$ and $\{T \geq n+1\} = \{T > n\} \in \mathcal{F}_n \subset \mathcal{F}_{n+1}$, hence $\{T = n+1\} = \{T \leq n+1\} \cap \{T \geq n+1\} \in \mathcal{F}_{n+1}$. On the other hand, $\{T = 0\} = \{T \leq 0\} = \{T > 0\}^c \in \mathcal{F}_0$. So $\{T = n\} \in \mathcal{F}_n$ for all $n \geq 0$.

□

Exercise 9.5.5.

Solution. Let $k \in \mathbb{N}$. Then $\{T = k\} \cap \{T = n\} = \emptyset \in \mathcal{F}_n$ if $k \neq n$, and $\{T = k\} \cap \{T = n\} = \{T = n\} \in \mathcal{F}_n$ if $k = n$. Hence $\{T = k\} \in \mathcal{F}_T$. So T is $\mathcal{F}_T$ -measurable.

□

Exercise 9.5.6.

Solution. Consider a filtration $\mathcal{F}_n = \sigma(S_0, S_1, \dots, S_n)$ as in **Exercise 9.5.1**. Define $T = \min\{n : S_n = 2\}$. Let $A = \{S_1 = 1, S_2 = 0, S_3 = 1, S_4 = 2\}$. Clearly A is a proper subset of $\{T = 4\}$, and hence $A \notin \sigma(T)$. But $A \cap \{T = j\} = \emptyset \in \mathcal{F}_n$ for all $j \neq 4$, and $A \cap \{T = 4\} = A \in \mathcal{F}_4$. Thus

$A \in \mathcal{F}_T \setminus \sigma(T)$. By the previous exercise we then know that $\sigma(T)$ is a proper subset of $\mathcal{F}_T$.

□

Exercise 9.5.7.

Solution. Let $n \geq 0$. Then $A \cap \{T_{a,b} = n\} = \{\text{visits } 0 \text{ exactly } k \text{ times by time } n\} \cap \{T_{a,b} = n\} \in \mathcal{F}_n$, since the first set is in $\sigma(S_0, \dots, S_n) = \mathcal{F}_n$. So by definition $A \in \mathcal{F}_T$.

□

Exercise 9.5.8.

Solution. Let $E \in \mathcal{B}_{\mathbb{R}}$. $X_T(\omega) \in E$ if and only if $T(\omega) = n$ for some n and $X_n(\omega) \in E$. Hence $\{X_T \in E\} = \cup_{n \geq 0} \{X_n \in E, T = n\}$. Therefore for any $k \geq 0$,

$$\{X_T \in E\} \cap \{T = k\} = \{X_k \in E, T = k\} \in \mathcal{F}_k.$$

So $\{X_T \in E\} \in \mathcal{F}_T$, i.e. X_T is $\mathcal{F}_T$ -measurable.

□

9.6.

Exercise 9.6.2.

Solution. Since $T \wedge n$ is a bounded stopping time, by optional stopping theorem we have $\mathbb{E}(S_{T \wedge n}^2 - T \wedge n) = \mathbb{E}(S_0^2 - 0) = 0$. So $\mathbb{E}(S_{T \wedge n}^2) = \mathbb{E}(T \wedge n)$. By DCT and MCT, $\lim_{n \rightarrow \infty} \mathbb{E}(S_{T \wedge n}^2) = \mathbb{E}(S_T^2) = \mathbb{E}(T) = \lim_{n \rightarrow \infty} \mathbb{E}(T \wedge n)$. Hence

$$\mathbb{E}(T) = \mathbb{E}(S_T^2) = a^2 \mathbb{P}(S_T = a) + b^2 \mathbb{P}(S_T = b) = \frac{a^2 b}{b - a} - \frac{b^2 a}{b - a}.$$

□

Exercise 9.6.7.

Solution. Since $\mathbb{E}(S_n - \mu n \mid \mathcal{F}_{n-1}) = \mathbb{E}X_n + S_{n-1} - \mu n = S_{n-1} - \mu(n-1)$, $S_n - \mu n$ is a martingale adapted to $\{\mathcal{F}_n\}$. Then by optional stopping theorem and $T \wedge n$ is a bounded stopping time, $\mathbb{E}(S_{T \wedge n} - \mu(T \wedge n)) = \mathbb{E}(S_0 - \mu \cdot 0) = 0$. Now by DCT and MCT we have $\mathbb{E}(S_T) = \mu \mathbb{E}(T)$.

9.7.

Exercise 9.7.4.

Solution. For each point in $\{T > n\}$, X_n is an odd number which is not a prime. And $X_n \leq p_{n-1} + 2$

where p_{n-1} is the largest prime divisor of X_{n-1} . That is,

$$X_n \leq \frac{X_{n-1}}{3} + 2 \leq \frac{X_{n-1}}{\sqrt{2}} \text{ a.s. on } \{T > n\}.$$

By monotonicity of conditional expectation and $\{T > n\} \in \mathcal{F}_{n-1}$,

$$\mathbb{E}(X_n \mathbf{1}_{\{T > n\}} | \mathcal{F}_{n-1}) \leq \mathbb{E}\left(\frac{X_{n-1}}{\sqrt{2}} \mathbf{1}_{\{T > n\}} \middle| \mathcal{F}_{n-1}\right) \leq \frac{X_{n-1}}{\sqrt{2}} \mathbf{1}_{\{T > n\}}.$$

Hence on $\{T > n\}$ a.s., by Jensen's inequality for concave function,

$$\mathbb{E}(\log(X_n) | \mathcal{F}_n) \leq \log(X_{n-1}) - \frac{1}{2} \log(2).$$

Since $\log(X_{T \wedge n}) \geq 0$, by the conclusion in (9.7.2) we have

$$\mathbb{E}(T) \leq \frac{\log(x) - 0}{\frac{1}{2} \log(2)} = C \log(x).$$

□

Solution. Write $S_n = X_1 + \cdots + X_n$. For any w_m (they're measurable),

$$\mathbb{E}w_m(x + X) \big|_{x=S_n} = \mathbb{E}_{X_{n+1}} w_m(S_n + X_{n+1}) = \mathbb{E}(w_m(S_{n+1}) | \mathcal{F}_n),$$

by the independence of S_n and X_{n+1} . So

$$\begin{aligned} w_{N-n}(S_n) &= (S_n - K)^+ \vee \mathbb{E}w_{N-n-1}(x + X) \big|_{x=S_n} \\ &= (S_n - K)^+ \vee \mathbb{E}_{X_{n+1}} w_{N-(n+1)}(S_{n+1}) \\ &= (S_n - K)^+ \vee \mathbb{E}(w_{N-(n+1)}(S_{n+1}) | \mathcal{F}_n), \end{aligned}$$

which means $w_{N-n}(S_n) \geq (S_n - K)^+$ and $\mathbb{E}(w_{N-(n+1)}(S_{n+1}) | \mathcal{F}_n) \leq w_{N-n}(S_n)$, therefore $w_{N-n}(S_n)$ is a Snell envelope. Hence the optimal τ would be

$$\tau = \min\{n : (S_n - K)^+ = w_{N-n}(S_n)\}.$$

$w_{N-n}(S_n)$ represents the optimal payoff estimated at time n . At time $n - 1$, to estimate the optimal payoff, we compare the current payoff and the average of optimal payoff estimated at time n .

□

For non-negative supermartingale,

$$(a - b)\mathbb{E}(U_m) \geq \sum_{n=1}^m \mathbb{E}(Z_n(Y_n - Y_{n-1})) \geq \sum_{n=1}^m \mathbb{E}(Y_n - Y_{n-1})$$

9.9

$\{n-1, n\}$ -in-upcrossing indicator Z_n :

1. $Z_n = 1$: $n-1$ is in an upcrossing and $Y_{n-1} < b$.
2. $Z_n = 0$: Either (a) $n-1$ is in an upcrossing and $Y_{n-1} \geq b$, or (b) $n-1$ is not in an upcrossing.

So Z_n is $\mathcal{F}_{n-1}$ -measurable.

9.11

Exercise 9.11.1. Let $A \in \mathcal{E}$. Then since $A \in \mathcal{G} = \sigma(X_1, X_2, \dots)$, there exists some $B \subset \mathbb{R}^{\mathbb{N}}$ such that $A = \{(X_i)_{i \geq 1} \in B\}$. Given $\epsilon > 0$, since the algebra $\cup_{i \geq 1} \sigma(X_1, \dots, X_i)$ generates $\mathcal{G}$, by the **approximation lemma** we know that there exists some $n \in \mathbb{N}$ and $A_n \in \sigma(X_1, \dots, X_n)$ such that $\mathbb{P}(A \Delta A_n) < \epsilon$. Note that we can also find some $B_n \subset \mathbb{R}^n$ such that $A_n = \{(X_1, \dots, X_n) \in B_n\}$.

Next, define random vector $(Y_i)_{i \geq 1} = h_n((X_i)_{i \geq 1})$, where

$$h_n : (x_1, x_2, \dots) \mapsto (x_{n+1}, \dots, x_{2n}, x_1, \dots, x_n, x_{2n+1}, \dots).$$

Then by the *i.i.d.* assumption of r.v.s $\{(X_i)_{i \geq 1}\}$, $(Y_i)_{i \geq 1}$ and $(X_i)_{i \geq 1}$ have the same law, hence

$$\begin{aligned} \mathbb{P}(A \Delta A_n) &= \mathbb{P}(\{(X_i)_{i \geq 1} \in B\} \Delta \{(X_i)_{i=1}^n \in B_n\}) \\ &= \mathbb{P}(\{(Y_i)_{i \geq 1} \in B\} \Delta \{(Y_i)_{i=1}^n \in B_n\}) \\ (A'_n := \{(Y_i)_{i=1}^n \in B_n\}) &= \mathbb{P}(A \Delta A'_n) < \epsilon. \end{aligned}$$

The last equality holds because $A \in \mathcal{E}$ and h_n is a $2n$ -permutation,

$$\{(Y_i)_{i \geq 1} \in B\} = \{(X_i)_{i \geq 1} \in B\} = A.$$

So we now have both $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$ and $\mathbb{P}(A'_n) \rightarrow \mathbb{P}(A)$, and hence

$$\mathbb{P}(A_n) \mathbb{P}(A'_n) \rightarrow \mathbb{P}(A)^2.$$

Moreover,

$$\begin{aligned} \mathbb{P}(A_n \Delta A'_n) &= \mathbb{E} |\mathbb{1}_{A_n} - \mathbb{1}_{A'_n}| \leq \mathbb{E} |\mathbb{1}_{A_n} - \mathbb{1}_A| + \mathbb{E} |\mathbb{1}_A - \mathbb{1}_{A'_n}| \\ &= \mathbb{P}(A_n \Delta A) + \mathbb{P}(A \Delta A'_n) < 2\epsilon, \end{aligned}$$

which implies $\mathbb{P}(A_n) - \mathbb{P}(A_n \cap A'_n) < 2\epsilon$ and hence, $|\mathbb{P}(A) - \mathbb{P}(A_n \cap A'_n)| < 3\epsilon$. So we have

$$\mathbb{P}(A_n \cap A'_n) \rightarrow \mathbb{P}(A).$$

But $\{X_i\}_{i \geq 1}$ are independent, hence the events $A_n \in \sigma((X_i)_{i=1}^n)$ and $A'_n \in \sigma((Y_i)_{i=1}^n) = \sigma((X_i)_{i=n+1}^{2n})$ are independent,

$$\mathbb{P}(A_n \cap A'_n) = \mathbb{P}(A_n) \mathbb{P}(A'_n) \rightarrow \mathbb{P}(A).$$

Therefore $\mathbb{P}(A)^2 = \mathbb{P}(A)$, implies $\mathbb{P}(A) = 0$ or 1 .

□

Recall. Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For any $A \in \mathcal{G} = \sigma(X_1, X_2, \dots)$ and any $\epsilon > 0$, show that there is some $n \geq 1$ and some $A_n \in \sigma(X_1, \dots, X_n)$ such that $\mathbb{P}(A \Delta A_n) < \epsilon$.

Proof. We set $\mathcal{F} = \bigcup_{n \geq 1} \sigma(X_1, \dots, X_n)$, and therefore $\mathcal{G} = \sigma(\mathcal{F})$. Next we check that $\mathcal{F}$ is an algebra.

1. Clearly $\Omega \in \mathcal{L}$.
2. Suppose $C \in \mathcal{F}$. Then $A \in \sigma(X_1, \dots, X_n)$ for some $n \geq 1$, therefore $C^c \in \sigma(X_1, \dots, X_n) \subset \mathcal{F}$.
3. Take any finite collection $A_1, \dots, A_m \in \mathcal{F}$, then $A_1, \dots, A_m \in \sigma(X_1, \dots, X_M)$ for some $M \geq 1$ and therefore $\cup_{i=1}^m A_i \in \mathcal{F}$.

Hence by **approximation lemma**, there exists $A_\epsilon \in \mathcal{F}$ such that $\mathbb{P}(A_\epsilon \Delta A) < \epsilon$.

Now since $\mathcal{F} \subseteq \mathcal{L}$, $\mathcal{G} = \sigma(\mathcal{F}) \subseteq \sigma(\mathcal{L})$. Therefore for any set $B \in \sigma(\mathcal{L})$ (also for any set in $\mathcal{G}$, of course), there exists $A \in \mathcal{L}$ such that $\mathbb{P}(A \Delta B) < \epsilon$.

□

9.13.

Exercise 9.13.4. Suppose $\{X_n\}_{n \geq 0}$ is a martingale that is uniformly bounded by some $M > 0$ in L^2 . Then for any $0 \leq n \leq m$, $\|X_m - X_n\|_2 \leq 2M$ and

$$\begin{aligned}\mathbb{E}(X_m - X_n)^2 &= \mathbb{E}[\mathbb{E}((X_m - X_n)^2 | \mathcal{F}_n)] \\ &= \mathbb{E}X_m^2 - \mathbb{E}[2X_n\mathbb{E}(X_m | \mathcal{F}_n) - X_n^2] \\ &= \mathbb{E}X_m^2 - \mathbb{E}X_n^2.\end{aligned}$$

From this, we can write

$$\mathbb{E}(X_m - X_n)^2 = \sum_{i=n}^{m-1} (\mathbb{E}X_{i+1}^2 - \mathbb{E}X_i^2) = \sum_{i=n}^{m-1} \mathbb{E}(X_{i+1} - X_i)^2,$$

which shows that $\|X_n - X_0\|_2$ is a non-decreasing sequence of n . Therefore together with the uniformly bounded condition,

$$\lim_{n \rightarrow \infty} \|X_n - X_0\|_2^2 = \sum_{i=0}^{\infty} \mathbb{E}(X_{i+1} - X_i)^2 < \infty.$$

This means $\|X_m - X_n\|_2^2 = \sum_{i=n}^{m-1} \mathbb{E}(X_{i+1} - X_i)^2 < \epsilon$ for any sufficiently large n, m , and hence X_n is Cauchy in L^2 . Then by the completeness of $\|\cdot\|_2$ in L^2 space, $X_n \rightarrow X$ in L^2 for some $X \in L^2$. $\square$

Exercise 9.13.6. Suppose $p > 1$. Since X_n is a martingale or a nonnegative submartingale, $|X_n|^p$ is a nonnegative submartingale. Suppose X_n is uniformly bounded in L^p . Then by *Theorem 9.13.3*, we have $\mathbb{E}|X_n - X|^p \rightarrow 0$ for some $X \in L_p$. Since $t \mapsto |t|^p$ is a convex function, for any $\lambda \in (0, 1)$,

$$|X_n|^p = \left| (1 - \lambda) \frac{X_n - X}{1 - \lambda} + \lambda \frac{X}{\lambda} \right|^p \leq (1 - \lambda)^{1-p} |X_n - X|^p + \lambda^{1-p} |X|^p. \quad (1)$$

Given any $\epsilon > 0$, take $\lambda = \left(1 + \frac{\epsilon}{\|X\|_p^p}\right)^{1/(1-p)} \in (0, 1)$, then by (1) we have

$$\begin{aligned}||X_n|^p - |X|^p| &\leq \epsilon \cdot \frac{|X|^p}{\|X\|_p^p} + \left[1 - \left(\frac{\|X\|_p^p}{\|X\|_p^p + \epsilon}\right)^{\frac{1}{p-1}}\right] |X_n - X|^p \\ &< \epsilon \cdot \frac{|X|^p}{\|X\|_p^p} + |X_n - X|^p.\end{aligned}$$

Take expectation on both sides,

$$\mathbb{E}||X_n|^p - |X|^p| \leq \epsilon + \|X_n - X\|_p^p.$$

By the L^p convergence of X_n , $\|X_n - X\|_p^p \rightarrow 0$ as $n \rightarrow \infty$. Thus

$$\lim_{n \rightarrow \infty} ||X_n|^p - |X|^p|_1 \leq \epsilon.$$

Since $\epsilon > 0$ is arbitrary, we have shown that $|X_n|^p$ converges in L^1 to $|X|^p$. Then it follows by *Proposition 8.3.5* that $|X_n|^p$ is uniformly integrable. □

Exercise 9.13.7. Write the process $\{X_n^2\}$ as

$$X_n^2 = X_n^2 - \langle X \rangle_n + \langle X \rangle_n$$

where $\{X_n^2 - \langle X \rangle_n\}_{n \geq 0}$ is a martingale, since

...

Then by Doob's inequality for non-negative sub-martingale,

$$\mathbb{E} \max_{i \leq n} (X_i^2) \leq 4\mathbb{E} X_n^2 = 4\mathbb{E} \langle X \rangle_n.$$

Apply MCT on both sides, then we obtained

$$\mathbb{E} \sup_{n \geq 0} X_n^2 \leq 4\mathbb{E} \lim_{n \rightarrow \infty} \langle X \rangle_n = 4\mathbb{E} \langle X \rangle_\infty. \quad (2)$$

To discuss the property on the set $\{\langle X \rangle_\infty < \infty\}$, we start it by looking at $\{\langle X \rangle_\infty \leq k\}$, where $k \in \mathbb{N}$. Consider a random time

$$T_k := \max\{n : \langle X \rangle_n \leq k\}.$$

Then T_k is a stopping time, since $\langle X \rangle_n$ is non-decreasing and predictable:

$$\{T_k = n\} = \{\langle X \rangle_n \leq k\} \cap \{\langle X \rangle_{n+1} > k\} \in \mathcal{F}_n.$$

Therefore $X_{n \wedge T_k}$ is also a submartingale. By (2),

$$\sup_{n \geq 0} \mathbb{E} X_{n \wedge T_k}^2 \leq \mathbb{E} \sup_{n \geq 0} X_{n \wedge T_k}^2 \leq 4k.$$

Hence, by the submartingale convergence theorem (*THEOREM 9.13.3*), $\{X_{n \wedge T_k}\}$ converges a.s. and in L^2 to some r.v. Notice that $X_{n \wedge T_k} = X_n$ on $\{T_k = \infty\}$. But $\{\langle X \rangle_\infty \leq k\} = \{T_k = \infty\}$, so X_n converges a.s. on $\{\langle X \rangle_\infty \leq k\}$. Since $k \in \mathbb{N}$ is arbitrary, we have X_n converges a.s. on $\{\langle X \rangle_\infty < \infty\}$. □

Exercise 9.13.8 (Galton–Watson branching process).

(2) Since M_n is a martingale, $\mathbb{E} M_n = \mathbb{E} M_1 = \mathbb{E} X_{0,1} < \infty$ for all $n \geq 1$, therefore by the submartingale convergence theorem (*Theorem 9.9.2*), $M_n \rightarrow M$ a.s. and $\mathbb{E}|M| < \infty$. In particular, if $\mu < 1$, then $Z_n/\mu^n \rightarrow 0$ implies $Z_n \rightarrow 0$ as $n \rightarrow \infty$.

(3) Suppose $\mu = 1$. Then $Z_n = M_n$, and hence Z_n converges a.s. by (2). Also, we set $p_0 =$

$\mathbb{P}(X_{n,i} = 0)$. Let $k, N \in \mathbb{N}$. Then by the definition of Z_n , we can write

$$\mathbb{P}\left(\bigcap_{n \geq N} \{Z_n = k\}\right) = \mathbb{P}\left(\{Z_N = k\} \cap \bigcap_{n > N} \left\{\sum_{i=1}^k X_{n-1,i} = k\right\}\right). \quad (3)$$

Note that Z_N is $\sigma(X_{0,1}, X_{1,1}, X_{1,2}, \dots, X_{N-1,1}, X_{N-1,2}, \dots)$ -measurable, and the n -th summation set is $\sigma(X_{n-1,1}, X_{n-1,2}, \dots)$ -measurable. So by the i.i.d. assumption of $X_{n,i}$, the RHS is

$$\begin{aligned} \mathbb{P}\left(\{Z_N = k\} \cap \bigcap_{n > N} \left\{\sum_{i=1}^k X_{n-1,i} = k\right\}\right) &= \mathbb{P}(Z_N = k) \cdot \prod_{n > N} \mathbb{P}\left(\sum_{i=1}^k X_{n-1,i} = k\right) \\ &= \mathbb{P}(Z_N = k) \cdot \prod_{n > N} \mathbb{P}\left(\sum_{i=1}^k X_{1,i} = k\right). \end{aligned}$$

But since

$$\begin{aligned} \mathbb{P}\left(\sum_{i=1}^k X_{1,i} = k\right) &\leq 1 - \mathbb{P}(X_{1,1} = 0, X_{1,2} = 0, \dots, X_{1,k} = 0) \\ &= 1 - \prod_{i=1}^k \mathbb{P}(X_{1,i} = 0) = 1 - p_0^k < 1, \end{aligned}$$

we find that the RHS of (3) is zero, therefore $\mathbb{P}\left(\bigcap_{n \geq N} \{Z_n = k\}\right) = 0$. Taking the union bounds, we then see that

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} Z_n = k\right) = \mathbb{P}\left(\bigcup_{N \geq 1} \bigcap_{n \geq N} \{Z_n = k\}\right) \leq \sum_{N \geq 1} \mathbb{P}\left(\bigcap_{n \geq N} \{Z_n = k\}\right) = 0.$$

Since $k \in \mathbb{N}$ is arbitrary, we've proven that $\mathbb{P}(\lim_{n \rightarrow \infty} Z_n = 0) = 1$. □

(4)

(5)

(6) By THEOREM 9.13.3, $M_n \rightarrow M$ a.s. and in L^2 , thus $M_n \rightarrow M$ in L^1 by Jenson's inequality. By Fatou's lemma,

$$\mathbb{E}M \leq \liminf_{n \rightarrow \infty} \mathbb{E}M_n \leq \limsup_{n \rightarrow \infty} \mathbb{E}M_n \leq \mathbb{E}M + \limsup_{n \rightarrow \infty} \mathbb{E}|M_n - M| = \mathbb{E}M.$$

Therefore $\mathbb{E}M = \lim_{n \rightarrow \infty} \mathbb{E}M_n = \mathbb{E}M_1 > 0$, so $\mathbb{P}(M > 0) > 0$. □

Exercise 9.13.9. Suppose $X \in L^p$ for some $p \geq 1$. Recall the setting in **Exercise 9.12.2** that with the filtration $\mathcal{F}_n = \sigma\left(\bigcup_{i=0}^{2^n-1} \left[\frac{i}{2^n}, \frac{i+1}{2^n}\right)\right)$, the process $X_n := \mathbb{E}(X|\mathcal{F}_n)$ is a martingale. Since $\mathbb{E}|X_n|^p = \mathbb{E}|X|^p < \infty$ for all $n \geq 1$, by the martingale L^p convergence theorem (*Theorem 9.13.3*) we can know that $X_n \rightarrow X$ in L^p .

□

Exercise 9.13.11.

- (2) **(Uniformly bounded)** Let $\phi_N(x) := \sqrt{N}(\cos \pi x)^{2N}$ for $N \geq 1$. Then for $0 < a < b < 1$, $x \in [0, 1]$,

$$f_N(x) = \int_{a-x}^{b-x} \phi_N(t) dt.$$

Claim 1. For any $N \geq 1$ and $x \in [0, 1]$,

$$f_N(x) \leq \int_{-\frac{b-a}{2}}^{\frac{b-a}{2}} \phi_N(t) dt = f_N\left(\frac{a+b}{2}\right).$$

Proof. Since $\phi_n(t) = \sqrt{n}(\cos \pi t)^{2n}$ is 1-periodic, $f_n(x) := \int_{a-x}^{b-x} \phi_n(t) dt$ is also 1-periodic. The derivative of f_n is

$$\frac{d}{dx} f_n(x) = -\phi_n(b-x) + \phi_n(a-x),$$

and we can easily see that

$$f'_n(x) = 0 \Leftrightarrow \frac{b+a}{2} - x = \frac{k}{2}, k \in \mathbb{Z}$$

by the property of ϕ_n . Also, by the 1-periodicity, the global maxima must be $f(\frac{b+a}{2} + k)$ or $f(\frac{b+a+1}{2} + k)$, $k \in \mathbb{Z}$. If $\Delta := b - a \leq \frac{1}{2}$, since $\phi_n(s) \geq \phi_n(t)$ for all $|s| \leq \frac{\Delta}{2}$ and $|t - \frac{1}{2}| \leq \frac{\Delta}{2}$,

$$f\left(\frac{b+a-1}{2}\right) = \int_{\frac{1}{2}-\frac{\Delta}{2}}^{\frac{1}{2}+\frac{\Delta}{2}} \phi_n \leq \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} \phi_n = f\left(\frac{b+a}{2}\right).$$

If $\Delta > \frac{1}{2}$,

$$\begin{aligned} f\left(\frac{b+a}{2}\right) &= 2 \int_0^{\frac{\Delta}{2}} \phi_n = 2 \left(\int_{\frac{1}{2}-\frac{\Delta}{2}}^{\frac{\Delta}{2}} \phi_n + \int_0^{\frac{1}{2}-\frac{\Delta}{2}} \phi_n \right) \\ &\geq 2 \left(\int_{\frac{1}{2}-\frac{\Delta}{2}}^{\frac{\Delta}{2}} \phi_n + \int_{\frac{\Delta}{2}}^{\frac{1}{2}} \phi_n \right) = f\left(\frac{b+a-1}{2}\right). \end{aligned}$$

Therefore, $f(\frac{b+a}{2} + k)$, $k \in \mathbb{Z}$ is the maximum. Hence the claim follows. □

We keep using the notation $\Delta = b - a$ in the following, and abbreviate $f_n(\frac{a+b}{2})$ as f_n . First,

write

$$\begin{aligned}
\int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} (\cos \pi t)^{2n} dt &= \frac{1}{\pi} (\cos \pi t)^{2n-1} \sin \pi t \Big|_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} + (2n-1) \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} (\cos \pi t)^{2n-2} (\sin \pi t)^2 dt \\
&\Rightarrow 2n \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} (\cos \pi t)^{2n} dt = \frac{2}{\pi} \left(\cos \frac{\pi \Delta}{2} \right)^{2n-1} \sin \frac{\pi \Delta}{2} + (2n-1) \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} (\cos \pi t)^{2n-2} dt \\
&\Rightarrow f_n = \underbrace{\frac{1}{\pi \sqrt{n}} \left(\cos \frac{\pi \Delta}{2} \right)^{2n-1} \sin \frac{\pi \Delta}{2}}_{b_n} + \underbrace{\frac{2n-1}{2\sqrt{n(n-1)}}}_{c_n} f_{n-1} = b_n + c_n f_{n-1}.
\end{aligned}$$

Since $c_n > 1$ and $b_n > 0$ for all $n \geq 1$ (since $0 < \Delta < 1$), we can see that $\{f_n\}$ is a nonnegative monotonically increasing sequence. Moreover,

$$f_n = b_n + c_n b_{n-1} + c_n c_{n-1} b_{n-2} + \cdots + (c_n c_{n-1} \cdots c_3) b_2 + (c_n c_{n-1} \cdots c_2) f_1. \quad (4)$$

Claim 2. The product $\prod_{k=n}^m c_k$ is bounded above and below by some constants for all $2 \leq n \leq m$.

Proof. Clearly 1 is an lower bound since $c_n > 1$ for all $n \geq 1$. For the upper bound, we evaluate

$$\prod_{i=2}^n c_i = \prod_{i=2}^n \frac{2i-1}{2\sqrt{i(i-1)}} = \prod_{i=2}^n \sqrt{\frac{i}{i-1}} \frac{2i-1}{2i} = \sqrt{n} \prod_{i=2}^n \frac{2i-1}{2i}.$$

Note that the sequence $(1 - \frac{1}{2n}) \uparrow 1$ as $n \rightarrow \infty$, so we have the following bound:

$$\log \prod_{i=2}^n \left(1 - \frac{1}{2i}\right) = \sum_{i=2}^n \log \left(1 - \frac{1}{2i}\right) \leq \int_1^{n+1} \log \left(1 - \frac{1}{2x}\right) dx.$$

The right hand side is calculated as

$$\begin{aligned}
\int_1^{n+1} \log \left(1 - \frac{1}{2x}\right) dx &= \int_1^{n+1} \log(2x-1) - \log(2x) dx \\
&= \frac{1}{2} \left[(u \log u - u) \Big|_1^{2n+1} - (u \log u - u) \Big|_2^{2n+2} \right] \\
&= \frac{1}{2} \left(\log \frac{(2n+1)^{2n+1}}{(2n+2)^{2n+2}} + 2 \log 2 \right) \\
&= \log \sqrt{\frac{4(2n+1)^{2n+1}}{(2n+2)^{2n+2}}}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\sqrt{n} \prod_{i=2}^n \frac{2i-1}{2i} &\leq \sqrt{\frac{4n(2n+1)^{2n+1}}{(2n+2)^{2n+2}}} = \sqrt{\frac{4n}{2n+1}} \sqrt{\left(1 - \frac{1}{2n+2}\right)^{2n+2}} \\
&= \sqrt{2 \left(1 - \frac{1}{2n+1}\right)} \sqrt{\left(1 - \frac{1}{2n+2}\right)^{2n+2}} \uparrow \frac{2}{\sqrt{e}}
\end{aligned} \quad (5)$$

as $n \rightarrow \infty$. Hence $\frac{2}{\sqrt{e}}$ is an upper bound. $\square$

Now by (4) and (5) we have

$$f_n \leq \frac{2}{\sqrt{e}} \left(\sum_{i=2}^n b_i + f_1 \right),$$

where

$$\sum_{i=2}^n b_i = \sum_{i=2}^n \frac{1}{\pi \sqrt{i}} \left(\cos \frac{\pi \Delta}{2} \right)^{2i-1} \sin \frac{\pi \Delta}{2} = \frac{1}{\pi} \tan \frac{\pi \Delta}{2} \sum_{i=2}^n \frac{1}{\sqrt{i}} \left(\cos \frac{\pi \Delta}{2} \right)^{2i}.$$

Notice that the sequence $\frac{1}{\sqrt{n}} \left(\cos \frac{\pi \Delta}{2} \right)^{2n} \downarrow 0$ as $n \rightarrow \infty$, hence we can write

$$\begin{aligned} \sum_{i=2}^n \frac{1}{\sqrt{i}} \left(\cos \frac{\pi \Delta}{2} \right)^{2i} &\leq \int_1^n \frac{1}{\sqrt{x}} \left(\cos \frac{\pi \Delta}{2} \right)^{2x} dx \\ &= \int_1^n x^{-\frac{1}{2}} \exp \left[2x \log \left(\cos \frac{\pi \Delta}{2} \right) \right] dx \\ &\leq \frac{\Gamma \left(\frac{1}{2} \right)}{\sqrt{2 \log \left(\frac{1}{\cos \frac{\pi \Delta}{2}} \right)}}. \end{aligned}$$

Where the last inequality is obtained by bounding via integrating on $(0, \infty)$. Finally, the upper bound of f_n is

$$f_n \left(\frac{a+b}{2} \right) \leq \frac{2}{\sqrt{e}} \left[\frac{\Gamma \left(\frac{1}{2} \right) \tan \frac{\pi \Delta}{2}}{\pi \sqrt{2 \log \left(\frac{1}{\cos \frac{\pi \Delta}{2}} \right)}} + \frac{1}{2} \left(\Delta + \frac{1}{\pi} \sin \pi \Delta \right) \right] < \infty$$

for all $n \geq 1$. Since we've already know that $f_n \left(\frac{a+b}{2} \right)$ is increasing, $f_n \left(\frac{a+b}{2} \right) \uparrow L$ for some $L < \infty$. Therefore, together with *Claim 1*, $f_n(x)$ is bounded by L for all $n \geq 1$ on $x \in [0, 1]$.

(Convergence)

$\square$

(3) By (2) we can see that $|g(x)f_N(x)| \leq c|g(x)|$ for all $N \geq 1$ on $[0, 1]$, hence by dominated convergence theorem,

$$\lim_{N \rightarrow \infty} \int g(x)f_N(x)dx = \int cg(x)\mathbf{1}_{(a,b)}dx = c \int_a^b g(x)dx. \quad (6)$$

Where $c\mathbb{1}_{(a,b)}(x) = \lim_{N \rightarrow \infty} f_N(x)$ a.e. on $x \in [0, 1]$. Now fix a $N \geq 1$,

$$\begin{aligned} \int_0^1 g(x) f_N(x) dx &= \int_0^1 \int_a^b g(x) \sqrt{N} \left(\frac{1 + \cos 2\pi(u-x)}{2} \right)^N du dx \\ &= \int_0^1 \int_a^b g(x) \sqrt{N} (\cos \pi(u-x))^{2N} du dx \\ &= \int_a^b \sqrt{N} \int_0^1 g(x) (\cos \pi(u-x))^{2N} dx du, \end{aligned} \quad (7)$$

where we expand $(\cos \pi(u-x))^{2N}$ as

$$\begin{aligned} (\cos \pi(u-x))^{2N} &= \left(\frac{e^{i\pi(u-x)} + e^{-i\pi(u-x)}}{2} \right)^{2N} \\ &= \frac{e^{-i2\pi N(u-x)}}{4^N} (e^{i2\pi(u-x)} + 1)^{2N} \\ &= \frac{e^{-i2\pi N(u-x)}}{4^N} \sum_{k=0}^{2N} \binom{2N}{k} e^{i2\pi k(u-x)} \\ &= \frac{1}{4^N} \sum_{k=-N}^N \binom{2N}{k+N} e^{i2\pi k(u-x)} \\ &= \frac{1}{4^N} \left[\binom{2N}{N} + 2 \sum_{k=1}^N \binom{2N}{k+N} \cos 2\pi k(u-x) \right]. \end{aligned}$$

Also note that $\cos 2\pi k(u-x) = \cos 2\pi k u \cos 2\pi k x + \sin 2\pi k u \sin 2\pi k x$. This implies that the integral

$$\int_0^1 g(x) (\cos \pi(u-x))^{2N} dx = 0,$$

since $(\cos \pi(u-x))^{2N}$ is a linear combination of $\{1, \cos 2\pi n x, \sin 2\pi n x\}_{n \geq 1}$ and g is orthogonal to them. Therefore by (7), $\int_0^1 g(x) f_N(x) dx = 0$. Thus by (6),

$$\int_a^b g(x) dx = 0.$$

Since $0 < a < b < 1$ is arbitrary, we can know that if g_n is defined on $[0, 1)$ as

$$g_n(x) = \sum_{i=0}^{2^n-1} 2^n \left[\int_{\frac{i}{2^n}}^{\frac{i+1}{2^n}} g(u) du \right] \mathbb{1}_{\left[\frac{i}{2^n}, \frac{i+1}{2^n}\right)}(x),$$

then we have $g_n(x) = 0$ on $[0, 1)$. Also, **Exercise 9.12.2** has shown that $g_n \rightarrow g$ a.e. as $n \rightarrow \infty$. Therefore, $g = 0$ a.e. on $[0, 1]$.

□

9.14

Exercise 9.14.3.

(2) It's clear that for any $n \geq 0$,

$$\int_{a_n}^{a_{n+1}} \frac{1}{x^2} dx \geq \int_{a_n}^{a_{n+1}} \frac{1}{a_{n+1}^2} dx.$$

Therefore

$$\sum_{n \geq 0} \frac{a_{n+1} - a_n}{a_{n+1}^2} \leq \sum_{n \geq 0} \int_{a_n}^{a_{n+1}} \frac{1}{x^2} dx = \lim_{n \rightarrow \infty} \int_{a_0}^{a_n} \frac{1}{x^2} dx \leq \frac{1}{a_0} < \infty.$$

(1) First of all, if $a_n \uparrow A < \infty$, then similar to the above,

$$\sum_{n \geq 0} \frac{a_{n+1} - a_n}{a_n} \leq \lim_{n \rightarrow \infty} \int_{a_0}^{a_n} \frac{1}{x} dx = A - \log(a_0) < \infty.$$

If $a_n \uparrow \infty$, we discuss under two cases. Suppose $\liminf a_n/a_{n+1} = 0$. Then there is a subsequence $\{a_{n_k}\}_{k \geq 1}$ such that $a_{n_k}/a_{n_k+1} \rightarrow 0$ as $k \rightarrow \infty$. Hence for an $\epsilon \in (0, 1)$, there exists an $N \geq 1$ such that

$$\sum_{n \geq 0} \left(1 - \frac{a_n}{a_{n+1}}\right) \geq \sum_{k > N} \left(1 - \frac{a_{n_k}}{a_{n_k+1}}\right) \geq \sum_{k > N} (1 - \epsilon) = \infty.$$

Suppose $\liminf a_n/a_{n+1} = c > 0$. Then we can take an $N \geq 1$ such that

$$\frac{c}{2} < \inf_{n > N} \left(\frac{a_n}{a_{n+1}}\right) \leq c.$$

Define a number M as

$$M := \min_{0 \leq n \leq N} \left(\frac{a_n}{a_{n+1}}\right) \wedge \frac{c}{2},$$

then $0 < M \leq \frac{a_n}{a_{n+1}}$ for all $n \geq 0$. Therefore

$$\begin{aligned} \sum_{n \geq 0} \frac{a_{n+1} - a_n}{a_{n+1}} &\geq M \sum_{n \geq 0} \frac{a_{n+1} - a_n}{a_n} \\ &\geq M \sum_{n \geq 0} \int_{a_n}^{a_{n+1}} \frac{1}{x} dx = M \left[\lim_{n \rightarrow \infty} \log(a_n) - \log(a_0) \right] = \infty. \end{aligned}$$

□

Exercise 9.14.4. Let $Z_n = S_n^2/c_n^2$. Since c_n is $\mathcal{F}_{n-1}$ -measurable, and S_n is square-integrable mar-

tingale,

$$\begin{aligned}
\mathbb{E}(Z_n | \mathcal{F}_{n-1}) &= \frac{1}{c_n^2} \mathbb{E}(S_{n-1}^2 + S_n^2 - S_{n-1}^2 | \mathcal{F}_{n-1}) \\
&= \frac{S_{n-1}^2}{c_{n-1}^2} + \frac{\mathbb{E}((S_n - S_{n-1})^2 | \mathcal{F}_{n-1})}{c_n^2} - \left(1 - \frac{c_{n-1}^2}{c_n^2}\right) \frac{S_{n-1}^2}{c_{n-1}^2} \\
&= Z_{n-1} + \frac{\sigma_n^2}{c_n^2} - \left(1 - \frac{c_{n-1}^2}{c_n^2}\right) \frac{S_{n-1}^2}{c_{n-1}^2}.
\end{aligned}$$

Also, since $c_{n-1} \leq c_n$ for all $n \geq 1$, the last two terms are nonnegative and $\mathcal{F}_{n-1}$ -measurable. Hence Z_n is a nonnegative almost supermartingale. By Robbins-Siegmund's theorem, we know that $Z_n \rightarrow Z$ a.s. for some Z that is finite a.s. and

$$\sum_{n=1}^{\infty} \left(\frac{c_n^2 - c_{n-1}^2}{c_n^2} \right) \frac{S_{n-1}^2}{c_{n-1}^2} < \infty$$

on the set $\{\sum_{n=1}^{\infty} \sigma_n^2 / c_n^2 < \infty\}$. Moreover, since c_n^2 is positive and increasing, by **Exercise 9.14.3** we know that

$$\sum_{n=1}^{\infty} \frac{c_n^2 - c_{n-1}^2}{c_n^2} = \infty.$$

Hence if $\sum_{n=1}^{\infty} \sigma_n^2 / c_n^2 < \infty$, we must have $S_n^2 / c_n^2 \rightarrow 0$, and thus $S_n / c_n \rightarrow 0$ as $n \rightarrow \infty$. □

Exercise 9.14.5. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ be the filtration. Suppose $S_n = \sum_{i=1}^n X_i X_{i-1}$. Then S_n is a $\mathcal{F}_n$ -adapted martingale, since by the i.i.d. and zero-mean condition of X_n ,

$$\mathbb{E}(S_n | \mathcal{F}_{n-1}) = S_{n-1} + \mathbb{E}(X_n) X_{n-1} = S_{n-1}.$$

Also,

$$\sigma_n^2 = \text{Var}(S_n | \mathcal{F}_{n-1}) = \mathbb{E}((S_n - S_{n-1})^2 | \mathcal{F}_{n-1}) = \mathbb{E}(X_n^2 X_{n-1}^2 | \mathcal{F}_{n-1}) = \sigma^2 X_{n-1}^2.$$

Here $\sigma^2 = \mathbb{E}X_0^2$ is a fixed constant. Then, by MCT,

$$\mathbb{E} \sum_{n \geq 1} \frac{\sigma_n^2}{n^2} = \sigma^2 \mathbb{E} \sum_{n \geq 1} \frac{X_{n-1}^2}{n^2} = \sigma^2 \sum_{n \geq 1} \frac{\mathbb{E}X_{n-1}^2}{n^2} = \sum_{n \geq 1} \frac{\sigma^4}{n^2} < \infty,$$

then we have $\sum_{n \geq 1} \sigma_n^2 / n^2 < \infty$ a.s. Hence by the previous exercise, we know that $S_n / n \rightarrow 0$ a.s. □

Exercise 9.14.6. Define the random variable $P_n := \sum_{k=0}^n p_k = \sum_{k=0}^n \mathbb{P}(X_k = 1 | \mathcal{F}_{k-1})$. We also define a sequence of nonnegative random variables $Z_n := (S_n - P_n)^2$. Then we consider the conditional expectation,

$$\mathbb{E} \left(\frac{Z_n}{P_n^2} \middle| \mathcal{F}_{n-1} \right) = \frac{1}{P_n^2} \mathbb{E}(Z_n - Z_{n-1} | \mathcal{F}_{n-1}) + \frac{Z_{n-1}}{P_{n-1}^2} - \left(1 - \frac{P_{n-1}^2}{P_n^2}\right) \frac{Z_{n-1}}{P_{n-1}^2}.$$

Rewrite the first term as

$$\begin{aligned}\mathbb{E}(Z_n - Z_{n-1} | \mathcal{F}_{n-1}) &= \mathbb{E}((S_n - P_n)^2 | \mathcal{F}_{n-1}) - (S_{n-1} - P_{n-1})^2 \\ &= \mathbb{E}(S_n^2 | \mathcal{F}_{n-1}) - 2P_n \mathbb{E}(S_n | \mathcal{F}_{n-1}) + P_n^2 - (S_{n-1} - P_{n-1})^2\end{aligned}$$

Notice that since $X_n = \mathbb{1}_{\{X_n=1\}} = (\mathbb{1}_{\{X_n=1\}})^2 = X_n^2$,

$$p_n = \mathbb{P}(X_n = 1 | \mathcal{F}_{n-1}) = \mathbb{E}(X_n | \mathcal{F}_{n-1}) = \mathbb{E}(X_n^2 | \mathcal{F}_{n-1}),$$

so we have

$$\begin{aligned}\mathbb{E}(S_n^2 | \mathcal{F}_{n-1}) &= S_{n-1}^2 + 2p_n S_{n-1} + p_n \\ 2P_n \mathbb{E}(S_n | \mathcal{F}_{n-1}) &= 2P_n(S_{n-1} + p_n) \\ \mathbb{E}(P_n^2 - (S_{n-1} - P_{n-1})^2 | \mathcal{F}_{n-1}) &= -S_{n-1}^2 + 2P_{n-1}S_{n-1} + 2P_{n-1}p_n + p_n^2\end{aligned}$$

Combine them together, we then get

$$\mathbb{E}(Z_n - Z_{n-1} | \mathcal{F}_{n-1}) = p_n - p_n^2 = p_n(1 - p_n) \leq p_n.$$

The last inequality holds since $0 \leq \mathbb{E}(\mathbb{1}_{\{X_n=1\}} | \mathcal{F}_{n-1}) \leq 1$. Therefore,

$$\mathbb{E}\left(\frac{Z_n}{P_n^2} \mid \mathcal{F}_{n-1}\right) \leq \frac{Z_{n-1}}{P_{n-1}^2} + \frac{p_n}{P_n^2} - \left(1 - \frac{P_{n-1}^2}{P_n^2}\right) \frac{Z_{n-1}}{P_{n-1}^2}.$$

Since Z_{n-1}/P_{n-1}^2 , p_n/P_n^2 are nonnegative and $\mathcal{F}_{n-1}$ -measurable, and $0 \leq P_{n-1} \leq P_n$, by Robbins-Siegmund's almost supermartingale theorem, we have $\left(\frac{S_n}{P_n} - 1\right)^2 \rightarrow L < \infty$ a.s. on $\left\{\sum_{n \geq 0} \frac{p_n}{P_n^2} < \infty\right\}$. Notice that by the first result of **Exercise 9.14.3**,

$$\sum_{n \geq 0} \frac{p_n}{P_n^2} = \sum_{n \geq 0} \frac{P_n - P_{n-1}}{P_n^2} < \infty \text{ a.s.}$$

And the second result,

$$\sum_{n \geq 0} \left(\frac{P_n^2 - P_{n-1}^2}{P_n^2}\right) = \infty,$$

We have $\left(\frac{S_n}{P_n} - 1\right)^2 \rightarrow 0$ a.s.

□

10.2

10.2.5. $\varphi(x) := \sqrt{x}$ on $[0, 1]$ with Lebesgue measure λ . □

Exercise 10.2.6. (Measure preserving) Let $\omega = (\omega_e)_{e \in E} \in \{0, 1\}^E$. Since μ is the law of the i.i.d. Bernoulli(p) random variables attached to each $e \in E$, the law of $(\omega_e)_{e \in E}$ and $(\omega_{x+e})_{e \in E}$ are the same, i.e. for all $A \subseteq \{0, 1\}^E$,

$$\mu((\omega_e)_{e \in E} \in A) = \mu((\omega_{x+e})_{e \in E} \in A).$$

Hence

$$\mu(T_x(\omega) \in A) = \mu((\omega_{x+e})_{e \in E} \in A) = \mu((\omega_e)_{e \in E} \in A).$$

(Ergodicity) Let $x \neq 0$. Let $(X_e)_{e \in E}$ be the random variables such that $X_e(\omega) = \omega_e$ for all $e \in E$. Then for any $k \geq 1$,

$$\{T_x^k \in A\} = \{(\omega_{kx+e})_{e \in E} \in A\} = \{(X_{kx+e}(\omega))_{e \in E} \in A\} \in \sigma((X_{kx+e})_{e \in E}).$$

Therefore

$$A \in \bigcap_{k \geq 1} \sigma((X_{kx+e})_{e \in E}) = \mathcal{T}.$$

Since $(X_e)_{e \in E}$ is i.i.d. Bernoulli(p) random variables under μ , by Kolmogorov's 0-1 law we have $\mu(A) \in \{0, 1\}$. Therefore T_x is an ergodic transformation. □

Exercise 10.2.7. Let $k \in \{0, 1, \dots\} \cup \{\infty\}$. Then for $\omega \in \Omega$, $N(\omega) = k$ if and only if $N(T_x(\omega)) = k$, i.e.

$$\{N = k\} = \{T_k \in \{N = k\}\}.$$

Hence $\{N = k\} \in \mathcal{I}$. Since T_k is ergodic, $\mathcal{I}$ is trivial. So N is a constant a.s. □

Exercise 10.2.8. Suppose $N = k$ almost surely for some $k > 1$. Consider a d dimensional cube $B_N = \{(x_1, \dots, x_d) \in \mathbb{Z}^d : -N \leq x_i \leq N\}$. Define

$$A_N := \{B_N \text{ intersects more than one } \infty \text{ cluster}\}.$$

Then

(1) $A_N \subseteq A_{N+1}$ for all $N \geq 1$.

(2) There exists some $N \geq 1$ s.t. $\omega \in A_N$ if and only if ω has more than one ∞ cluster.

Therefore

$$\lim_{N \rightarrow \infty} \mathbb{P}(A_N) = \mathbb{P}\left(\bigcup_{N \geq 1} A_N\right) = 1.$$

Hence there is some $N \geq 1$ such that $\mathbb{P}(A_N) > 0$. However, since A_N depends only on the edges

outside B_N , A_N is independent of $\sigma((X_e)_{e \in E(B_N)})$, so

$$\mathbb{P}(N_\infty = 1) \geq \mathbb{P}(A_N)p^{2(2N)(2N+1)} > 0,$$

which leads to a contradiction. □

Exercise 10.2.9. Define $u : [0, \frac{1}{2}] \rightarrow [0, 1]$ as $u(x) = 4x(1-x)$ for all $0 \leq x \leq \frac{1}{2}$. Then u is a bijection defined on $[0, \frac{1}{2}]$, and its inverse is $u^{-1}(t) = \frac{1-\sqrt{1-t}}{2}$ for all $t \in [0, 1]$.

$$\mathbb{P}(\varphi(x) \leq t) = \mathbb{P}(-4x(1-x) \geq -t) = \mathbb{P}\left(\left(x - \frac{1}{2}\right)^2 \geq \frac{1-t}{4}\right).$$

Since the density $\frac{1}{\pi\sqrt{x(1-x)}}$ is symmetric about $1/2$,

$$\mathbb{P}\left(\left(x - \frac{1}{2}\right)^2 \geq \frac{1-t}{4}\right) = 2\mathbb{P}\left(x \leq \frac{1}{2} - \sqrt{\frac{1-t}{4}}\right) = 2\mathbb{P}(x \leq u^{-1}(t)).$$

So

$$\frac{1}{2}\mathbb{P}(\varphi(x) \leq t) = \int_0^{u^{-1}(t)} \frac{1}{\pi\sqrt{x(1-x)}} dx.$$

Apply the change of variable $v = u(x)$ on the RHS, then it becomes

$$\int_0^{u(u^{-1}(t))} \frac{2}{\pi} \sqrt{\frac{1}{v}} \left| \frac{du^{-1}}{dv} \right| dv = \frac{1}{2\pi} \int_0^t \frac{1}{\sqrt{v(1-v)}} dv = \frac{1}{2}\mathbb{P}(x \leq t).$$

Therefore φ is a measure-preserving transform w.r.t. this $\mathbb{P}$. Since $\{[0, t) : 0 \leq t \leq 1\}$ generates $\mathcal{B}([0, 1])$, So $\mathbb{P}(\varphi(x) \in A) = \mathbb{P}(x \in A)$ for all $A \in \mathcal{B}([0, 1])$. □

Exercise 12.1.4.

Sol. (Separable) We claim that for any $f \in C[0, \infty)$, there is a sequence of functions of the following form,

$$p(x) = \sum_{j=0}^{\infty} p^{(j)}(x) \mathbf{1}_{[j, j+1]}(x),$$

where $p^{(j)}$ is the polynomial with rational coefficients, converge to f in ρ . Clearly, the set of functions of this form is countable since the cardinality is the same as $\mathbb{N}^2$ and $\mathbb{N}$.

Let $f \in C[0, \infty)$. By Weierstrass approximation theorem, for each $j \in \mathbb{Z}_{\geq 0}$, there is a sequence of polynomials with rational coefficients $(P_n^{(j)})_{n \geq 1}$ such that $\sup_{x \in [j, j+1]} |f(x) - P_n^{(j)}(x)| \rightarrow 0$ as $n \rightarrow \infty$. Define a sequence of functions

$$P_n(x) := \sum_{j=0}^{\infty} P_n^{(j)}(x) \mathbf{1}_{[j, j+1]}(x).$$

Then

$$\rho(f, P_n) = \sum_{j=0}^{\infty} 2^{-j} \frac{\|f - P_n\|_{[j, j+1]}}{1 + \|f - P_n\|_{[j, j+1]}} = \sum_{j=0}^{\infty} 2^{-j} \frac{\|f - P_n^{(j)}\|_{[j, j+1]}}{1 + \|f - P_n^{(j)}\|_{[j, j+1]}}.$$

Let $g_n : j \mapsto g_n(j) = \frac{\|f - P_n^{(j)}\|_{[j, j+1]}}{1 + \|f - P_n^{(j)}\|_{[j, j+1]}}$ be a sequence of function on $\mathbb{Z}_{\geq 0}$. Then $|g_n| \leq 1$, and $g_n(j) \rightarrow 0$ as $n \rightarrow \infty$ since $\|f - P_n^{(j)}\|_{[j, j+1]} \rightarrow 0$ as $n \rightarrow \infty$. By the dominated convergence theorem,

$$\lim_{n \rightarrow \infty} \rho(f, P_n) = \lim_{n \rightarrow \infty} \sum_{j=0}^{\infty} 2^{-j} g_n(j) = \sum_{j=0}^{\infty} 2^{-j} \lim_{n \rightarrow \infty} g_n(j) = 0.$$

Therefore $(C[0, \infty), \rho)$ is separable.

(Complete) Let $(f_n)_{n \geq 0}$ be a Cauchy sequence in $(C[0, \infty), \rho)$. Fix $k \in \mathbb{Z}_{\geq 0}$. Choose n, m sufficiently large so that $\rho(f_n, f_m) < 2^{-k} \frac{\varepsilon}{1 + \varepsilon}$, then

$$2^{-k} \frac{\|f_n - f_m\|_{[k, k+1]}}{1 + \|f_n - f_m\|_{[k, k+1]}} \leq \rho(f_n, f_m) < 2^{-k} \frac{\varepsilon}{1 + \varepsilon},$$

and we get

$$\|f_n - f_m\|_{[k, k+1]} \leq \varepsilon.$$

Therefore $(f_n|_{[k, k+1]})_{n \geq 1}$ is a Cauchy sequence in $C[k, k+1]$ equipped with the sup-metric, which is a complete metric space by **Exercise 12.1.3**. Moreover, the sup-metric limit of $f_n|_{[k, k+1]}$ is its pointwise limit $\lim_{n \rightarrow \infty} f_n|_{[k, k+1]}(x)$. This ensures that $f_n(x)$ converges for all $x \in \mathbb{R}_{\geq 0}$. By setting $f : x \mapsto f(x) := \lim_{n \rightarrow \infty} f_n(x)$, we know that f_n converges to f uniformly on any interval of the type $[j, j+1]$. It follows that f_n converges uniformly to f on any compact sets in $\mathbb{R}_{\geq 0}$.

Also, f is indeed the ρ -limit of f_n : By dominated convergence,

$$\rho(f_n, f) \rightarrow \sum_{j \geq 0} 2^{-j} \lim_{n \rightarrow \infty} \frac{\|f - f_n\|_{[j, j+1]}}{1 + \|f - f_n\|_{[j, j+1]}} = 0.$$

Finally, since for any $n \in \mathbb{N}$, $\delta > 0$,

$$|f(x + \delta) - f(x)| \leq |f(x + \delta) - f_n(x + \delta)| + |f_n(x + \delta) - f_n(x)| + |f_n(x) - f(x)|.$$

Then, we bound the RHS by choosing a n to bound $|f_n(x + s) - f(x + s)|$ for any $s \in [-\Delta, \Delta]$ by the truth of uniform convergence, and then choose $\delta < \Delta$ to bound $|f_n(x) - f_n(x + \delta)|$. This proves that $f \in C[0, \infty)$, and therefore the space $(C[0, \infty), \rho)$ is complete.